V. Kannnan
École Polytechnique

Open research position (Bachelor/M1/M2)
Mechanics of shape memory alloys under extreme dynamic environments

The Mechanics and Materials under eXtremes research group within the Laboratoire de Mécanique des Solides (LMS) at the Ecole Polytechnique in Palaiseau, France, is looking for passionate **Bachelor degree** students to study the **temperature-controlled phase transformations at high strain rates**. The project is expected to start between **January 11-18, 2027** for a duration of **8 weeks**.

Research topic

Context: Shape Memory Alloys (SMAs) are a class of active materials which exhibit "pseudo/super elastic" mechanical response, elasto/baro-caloric cooling, and mechanically-driven magnetic flux (for a special class of magnetic SMAs) — with applications ranging from impact damping to power generation. At the microscopic length scales, these phenomena occur due to spatially-heterogeneous structural changes in the atomic lattice — phase transformations — driven by mechanical, thermal and magnetic fields.

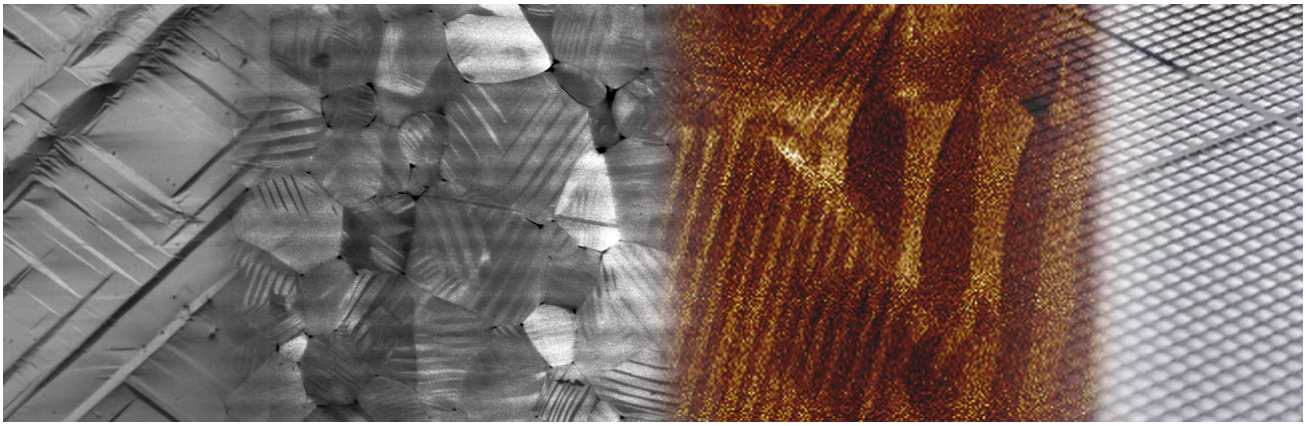
Research project: The student will design and implement upgrades to miniature Kolsky bar experiments with a focus on integrating multiple low and high speed actuation and instrumentation systems to work towards automatizing the existing experiments. Notably, these upgrades will enable in situ temperature control to study the macroscopic temperature dependent response of a single crystal CuAlNi SMA under high strain rates.

Your profile

- Third year undergraduate from the Math/Physics/Mechanics track of l'X, Master 1 in the mechanics/advanced materials track of IP Paris, or equivalent expertise.
- Background in mechanics and/or materials (demonstrated through coursework/internships)
- Eager to work on experimental development including mechanical testing and instrumentation
- Experience with data processing tools using Python/Matlab or equivalent
- Comfortable communicating in English, and enjoy working in an international environment.

Finally and most importantly, if you are passionate and eager to study mechanics, develop rigorous experiments, and enjoy spending many hours in an experimental lab, you are eligible to apply!

Application details in the next page.



Prof. Vignesh Kannan
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About our team – Mechanics and Materials under eXtremes

The mXm team, led by Prof. Vignesh Kannan, studies the mechanics of materials and structures under large stresses, multi-physical fields and short time scales. We use mechanical design, electronic and optical instrumentation to push the cutting edge of experimental mechanics down to nano-second time scales, and μm length scales. Our experiments are motivated by theoretical foundations, with the goal of understanding fundamental mechanisms that govern the multi-physical response of materials. We continuously seek to develop a collaborative, supportive and international group of scientists who enjoy the pursuit of exciting problems, and each other's company!

To apply!

Please send an email to Prof. Vignesh Kannan (vignesh.kannan@polytechnique.edu) with your CV and transcript with the subject **"Internship application – ..."**.